PREPRINT

# Automated protein design to disrupt disease-relevant protein–protein interactions

A framework of LLM agent skills with access to protein design software, backed by a literature database of digested scientific papers. A one-sentence prompt in; ranked, in silico validated binder designs out.

Michael Uckelmann
Hubrecht Institute, Utrecht, The Netherlands

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Little Protein Tiger (LPT) · github.com/MichaUckelmann/little-protein-tiger

FRAMEWORK

# Language-model reasoning, deterministic tools, and a curated literature layer.

01 **Pathway and target selection**
LLM skill queries the literature database and interaction graph; proposes a target pair.

02 **Structure and epitope analysis**
Candidate complexes ranked on measured interfaces; hotspots chosen and checked against the coordinates.

03 **Backbone generation**
RFD3 diffusion, sidecar prefilter, solubleMPNN sequence redesign.

04 **Refolding and gating**
RoseTTAFold3 refolds scored on ipTM, docking and fold RMSD, pLDDT, interface PAE, hotspot engagement.

05 **Ranking and report**
Composite z-score with ipTM and ipSAE; optional Rosetta terms.

Scale is set from a calibration run that measures the hit rate on its own output before production is sized.

LITERATURE LAYER

# Each paper is digested into a structured fingerprint.

Key findings with evidence, source span and confidence; proteins and structures; pathway context. Example (abridged):

## USP26 functions as a negative regulator of cellular reprogramming by stabilising PRC1 complex components

DOI 10.1038/s41467-017-00301-4 · pathway_biology · experimental_in_vitro · schema 2.0

CONTEXT HOOK

In mouse embryonic fibroblasts undergoing OSKM-mediated reprogramming, the deubiquitinase USP26 physically interacts with CBX4 and CBX6, removes K48-linked polyubiquitin and protects them from proteasomal degradation. Knockdown of Usp26 increases AP+ iPSC colony generation fourfold.

KEY FINDING 1 OF 5

| | |
|---|---|
| claim | shRNA knockdown of Usp26 enhances reprogramming efficiency fourfold |
| protein_pair | USP26 ↔ NANOG |
| evidence | ~400 AP+ colonies vs ~100 in control shRNA (quantitative) |
| context | Dox-induced OSKM MEF reprogramming, day 12 |
| confidence | 0.95 |
| source_span | Page 3, Para 1 |

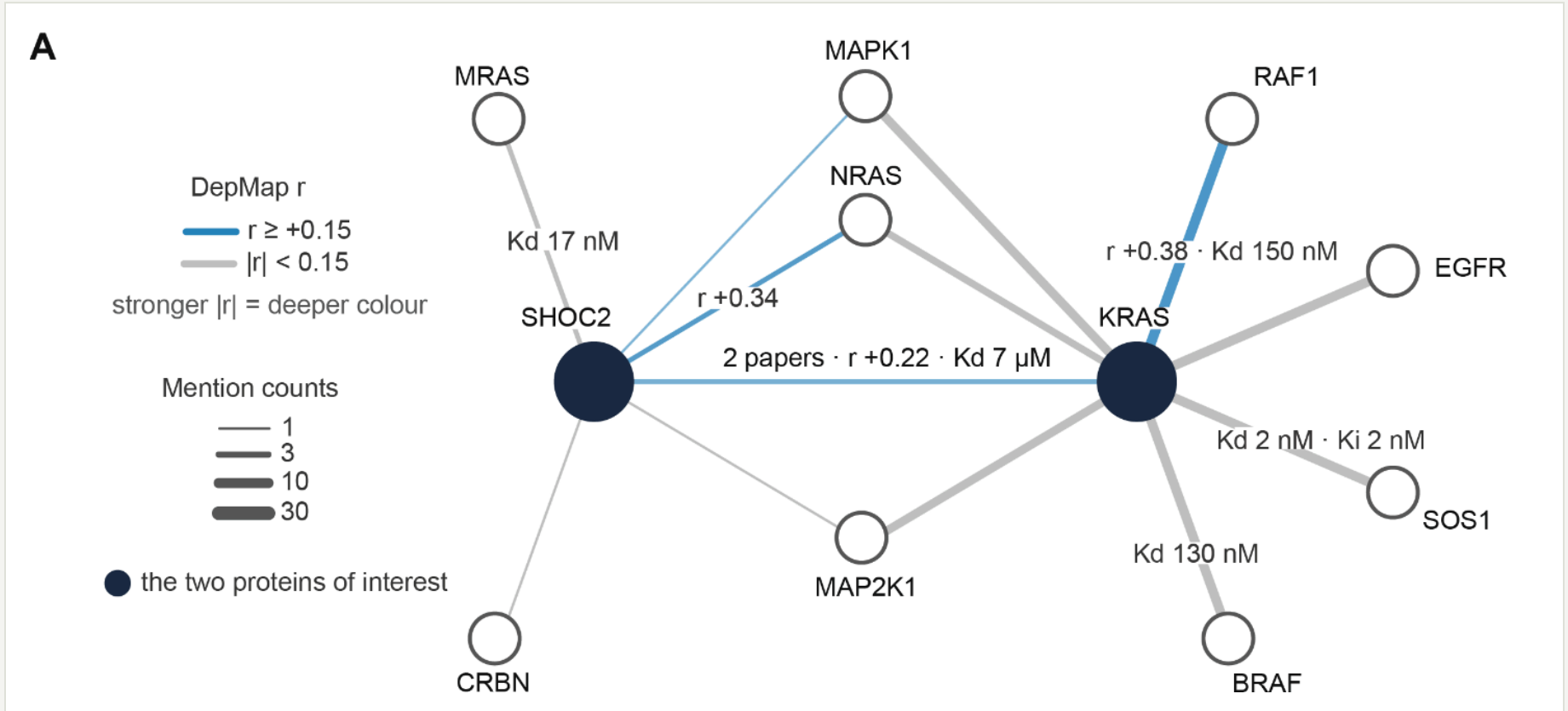
KEY FINDING 3 OF 5

| | |
|---|---|
| protein_pair | USP26 ↔ CBX6 |
| residues | Cys304 |
| evidence | Loss of K48 deubiquitination in C304S and Δ295–312 mutants (qualitative) |
| source_span | Page 5, Para 1 |

Fingerprints are embedded (PubMedBERT) for semantic search and aggregated into an interaction graph. Three complete examples: Supplemental Information 4.

FIGURE 2A / INTERACTION GRAPH

# Protein pairs from the corpus form a graph the skills can traverse.

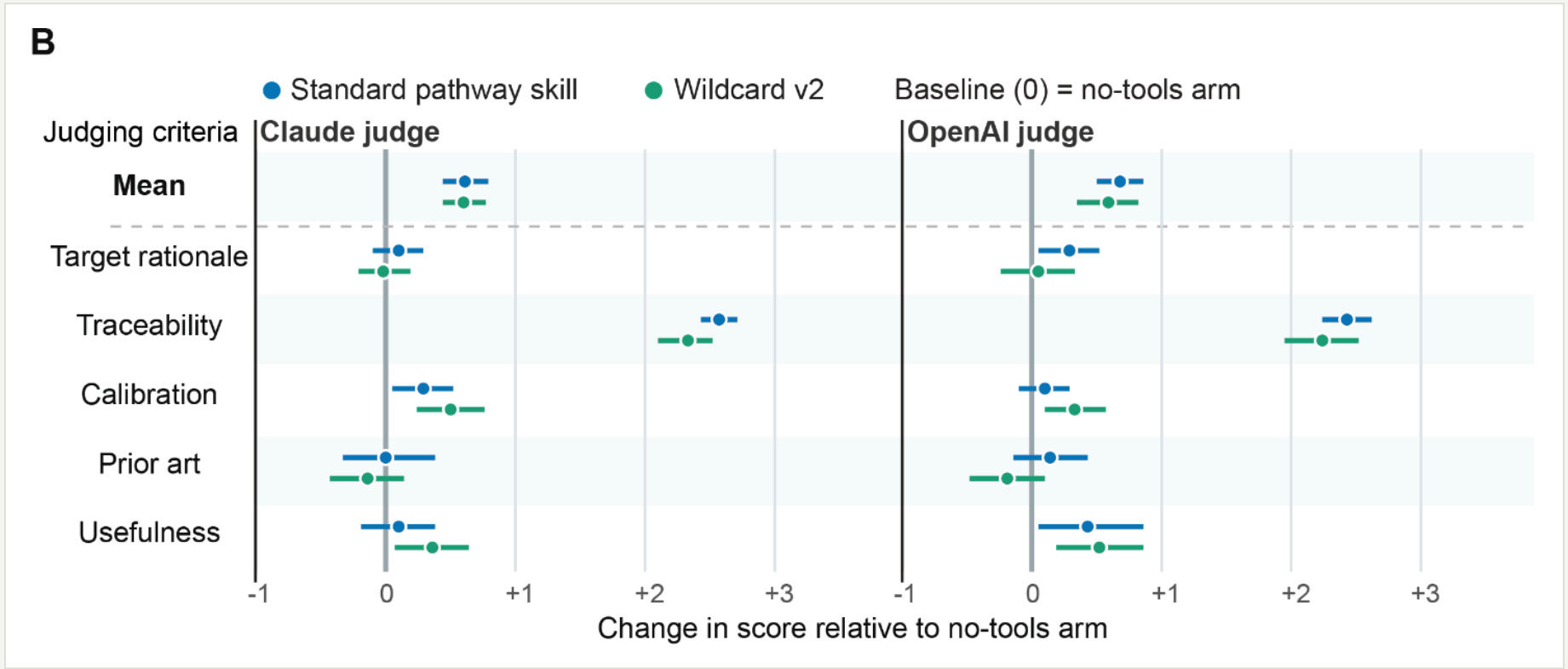



Nodes are proteins; edges carry affinity information and the number of papers mentioning the interaction (edge width). Edge colour marks the DepMap CRISPR gene-effect correlation of the pair (blue: r ≥ 0.15; grey: |r| < 0.15). The graph shown is a small subgraph.

The skills can query this graph, find shortest paths between proteins, and reason along edges with quantitative evidence attached.

FIGURE 2B / ABLATION

# Literature access mainly improves traceability.



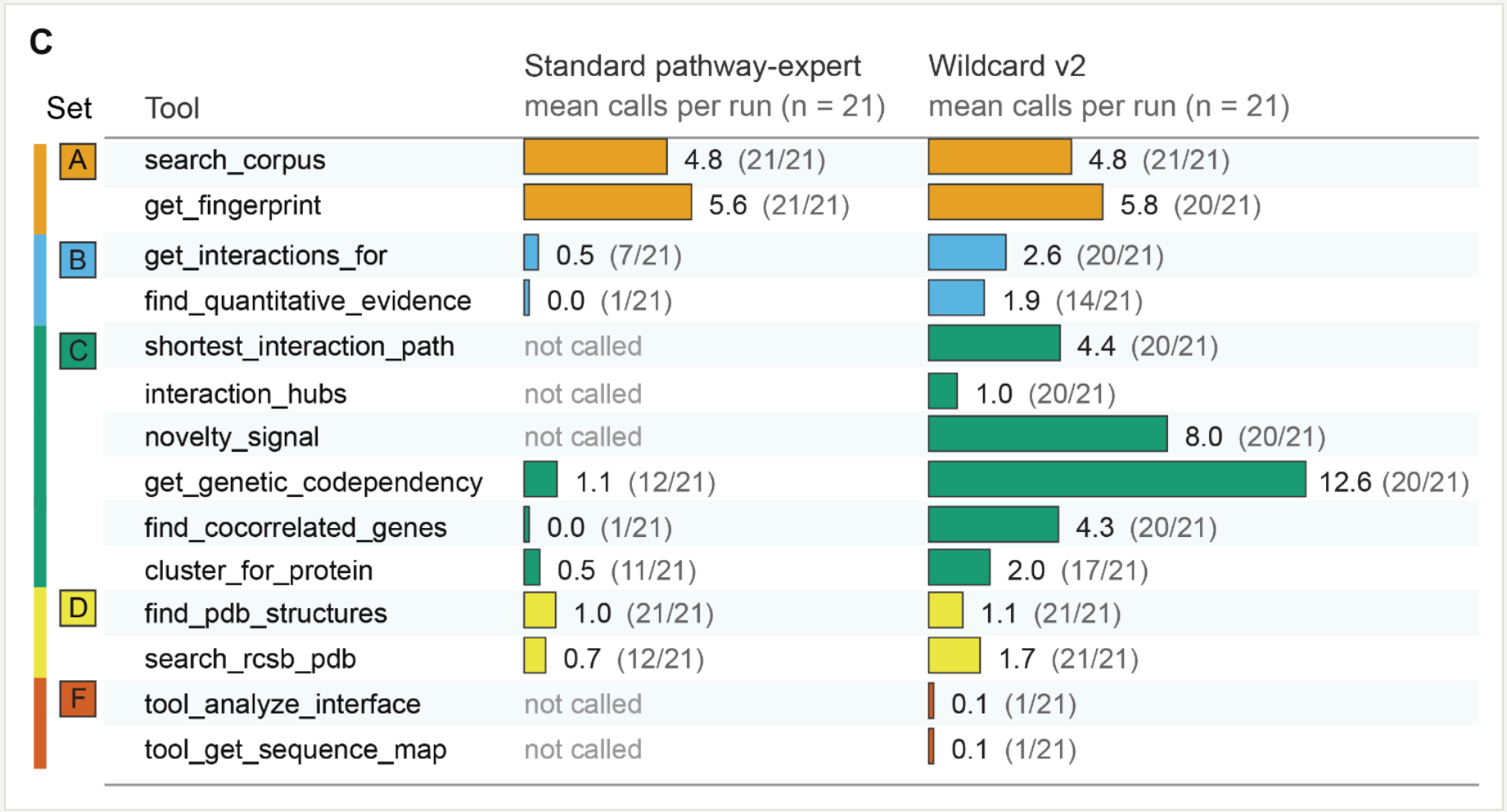

Rubric score (1–5) relative to a no-tools run; standard pathway skill and wildcard skill. Seven prompts, three replicates, 21 reports per arm; judged by claude-sonnet-5 and gpt-5.6-terra. Bars: 95% bootstrap interval over prompts.

READING

Traceability shows the largest gain for both skills and both judges. On target rationale, calibration, prior art and usefulness the differences are small and mostly within about half a point. The study does not assess whether the chosen targets are better, and the judging models may themselves be steered by their training knowledge.

FIGURE 2C / TOOL USE

# The wildcard skill uses the graph tools; the standard skill mostly does not.



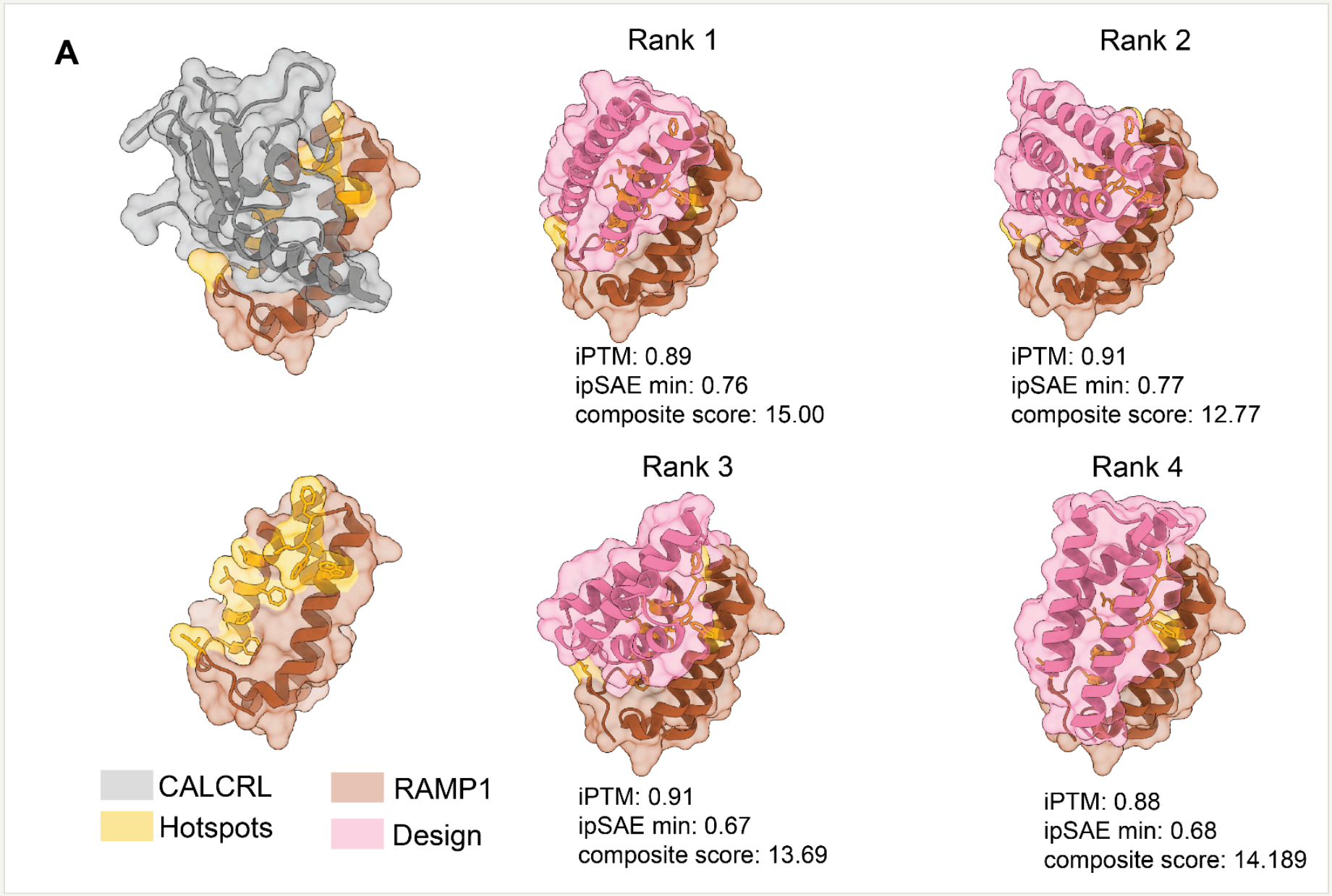

Mean tool calls per run (n = 21 per skill; fraction of runs calling the tool in brackets).

CLOSED-BOOK AGAINST OPEN-BOOK EXAM

48 detailed questions answerable from the database. Open book: 41 correct, 7 “unknown”. Closed book: 0 correct, 47 “unknown”, 1 wrong. The exam was biased towards database content and some questions were leading (Appendix II).

FIGURE 3A / END-TO-END RUN 1

# “Design novel inhibitors for pain receptors that are promising drug targets.”

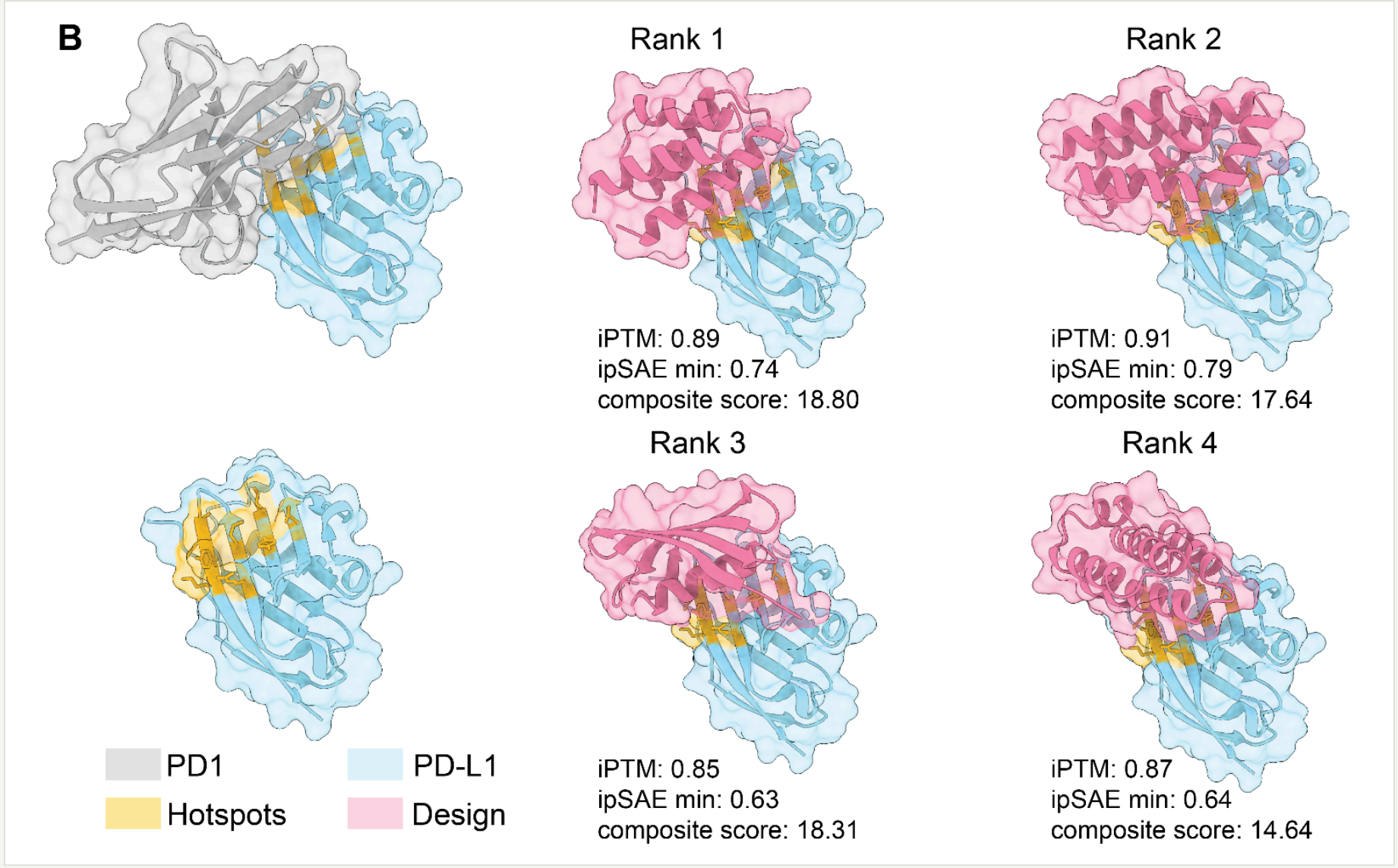



Selected target: CALCRL/RAMP1 (PDB 3N7S); hotspots (gold) on the RAMP1 heterodimerisation surface. Four highest-ranking designs (pink) with ipTM, minimum ipSAE and composite score.

17.9% of designs passed all hard gates and ipTM ≥ 0.7. All stage reports: Supplemental Information 5.

FIGURE 3B / END-TO-END RUN 2

# “Design immune checkpoint inhibitors.”




Selected target: PD-L1 in complex with PD-1 (PDB 4ZQK); hotspots (gold) on the PD-1-binding face. Four highest-ranking designs (pink) with ipTM, minimum ipSAE and composite score.

9.6% of designs passed all hard gates and ipTM ≥ 0.7. All stage reports: Supplemental Information 2.

SCOPE AND AVAILABILITY

# In silico results, at low cost.

**$0.33 / $0.42**

LLM API COST, RUNS 1 / 2

**~11 h / ~34 h**

GPU TIME, LOCAL WORKSTATION

LIMITATIONS STATED IN THE PREPRINT

Experimental success rates have not been evaluated; in silico metrics are a proxy. The corpus (about 14,000 papers in the example runs) limits target choice, and the released corpus is smaller because of licensing. There is no ground truth for whether a chosen target is a good one.

AVAILABILITY

Preprint with Supplemental Information 1–6 and Appendices I–II, code, skills and corpus tooling:

`github.com/MichaUckelmann/little-protein-tiger`